

Acoustic Feature-Based Detection and Classification of Speech Disfluencies Using Machine Learning

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Abstract:

Stutter identification and classification can be a labor-intensive process and may have subjectivity issues with what a speech-language pathologist classifies as a stutter. This paper investigates the effectiveness of traditional machine learning models in stuttering detection and classification. By extracting audio features from the UCLASS Stuttered Speech Clips dataset such as MFCCs, we can convert audio into a numeric representation a machine learning model can make classifications from. The results show that expanding from 13 MFCC features to a 29-feature vector and implementing a classifier chain leads to the F1 scores for the categories Block, Prolongation, SoundRep, WordRep, Interjection, and NoStutteredWords to be 0.79, 0.46, 0.80, 0.72, 0.66, and 0.92 and using random forest for binary classification leads to an F1 score of 0.93. A CNN trained on mel spectrograms underperformed traditional methods, suggesting deep learning requires substantially larger datasets to be effective in this domain. These results suggest that machine learning shows promise as a supplementary tool for speech-language pathologists in stuttering assessment.

Introduction:

Stuttering is a speech disorder that affects the fluency of one's speech, with more than 80 million people affected by it worldwide (1). The two main types include developmental stuttering, which occurs in children and typically resolves itself in childhood, and neurogenic stuttering, which can occur after brain trauma (3). Currently, treatment relies on manual assessment by a speech-language pathologist (SLP) to determine what constitutes a stutter as well as what type of stutter occurred. The limitations with this process are that it is a time consuming process, as manually counting and categorizing disfluency events across a full session is labor intensive and difficult to scale, along with the fact that it is a subjective matter. An SLP may assume something is a stutter even if it is not (3).

Machine learning shows promise, as it has the capabilities to process large amounts of audio data and objectively determine whether a stutter occurred and classify its type. Machine learning and deep learning methods have already been applied to speech-processing tasks, demonstrating strong performance in speech recognition (4). Several studies have previously explored ML's applications to stuttering detection and have shown promising results, with recent ones achieving F1 scores between 95% and 98% by using datasets of approximately 32,000 total clips (2). Less is known about optimizing traditional machine learning methods for smaller datasets, where deep learning approaches are prone to overfitting due to insufficient training samples (2).

In this study, we developed and analyzed a machine learning pipeline for detecting and classifying speech disfluencies using features extracted from UCLASS Stuttered Speech Clips (SEP-28k Format) (5). We hypothesized that acoustic features extracted from short audio clips could accurately detect and classify speech disfluencies, and that adding more features would improve classification compared to MFCCs alone.

Methods:

We used the UClass Stuttered Speech Clips dataset. This dataset comes pre-segmented into 3-second clips with binary labels for six disfluency types: Block, Prolongation, Sound Repetition, Word Repetition, Interjection, and NoStutteredWords. We extracted 13 Mel-Frequency Cepstral Coefficients (MFCCs) per clip as they are the standard feature for speech processing because of their ability to compress a spectrogram into a numeric representation. This was used as the baseline, and we later expanded to include 13 MFCC standard deviations to measure how much the means vary over time, zero crossing rate to aid in detection for stutter types such as blocks, root mean square energy to measure average loudness, and spectral centroid to further help distinguish stutter types. For binary classification, we compared logistic regression, support vector machine (SVM), and random forest. Due to random forest having the highest F1 score, that was used for classification per type of stutter. For multi-label classification, a classifier chain was implemented so that each classifier receives all 29 features alongside the predictions

of the preceding classifiers in the chain to account for clips that had multiple labels. A convolutional neural network (CNN) was also used to assess whether deep learning would outperform traditional methods on this dataset. Initial inspection revealed that the dataset was partially ordered by clip type, so the data was shuffled prior to splitting into testing and training groups using an 80/20 train-test split. The variable `random_state` was set to 42 to ensure reproducibility. Figures 1 and 2 show the waveform and mel spectrogram of a representative disfluent clip.

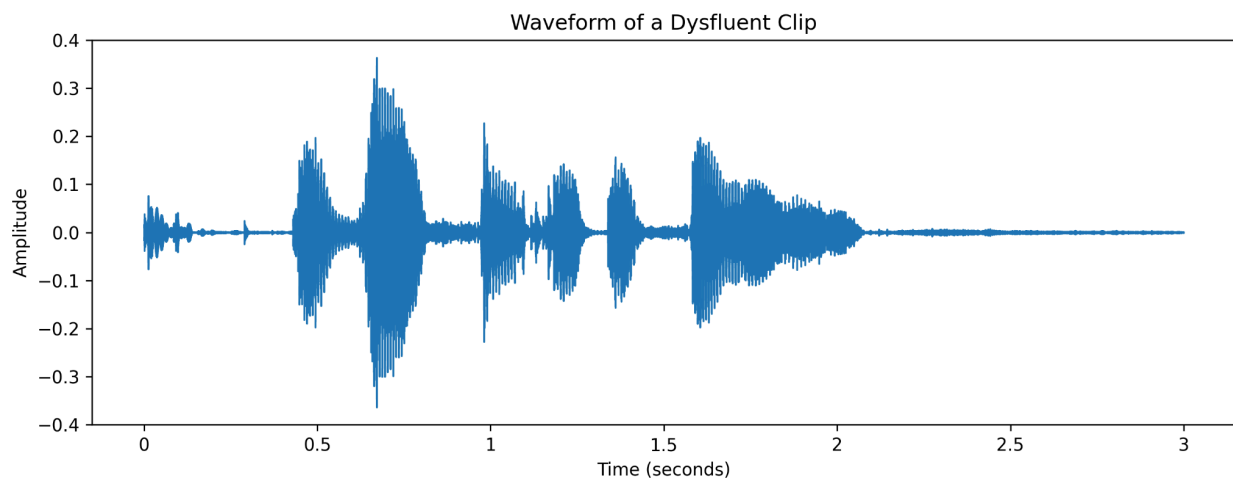


Figure 1. Waveform of disfluent clip showing a block from 0-0.3s, burst release from 0.5-0.7s, and sound repetitions from 1-2s

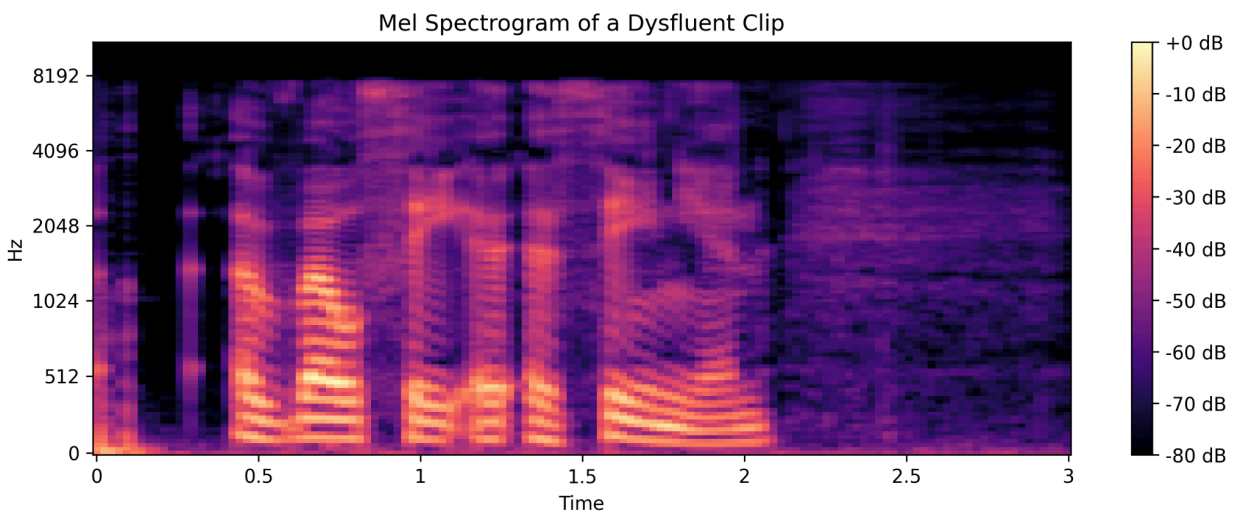


Figure 2. Mel Spectrogram showing the same disfluent clip. Thick, dark vertical stripes indicate blocks and repeating vertical stripes indicate repetition.

Results:

The results of binary classification show that random forest obtained the highest accuracy and F1 score, 92.8% and 0.93, followed by SVM, 86.4% and 0.87, with logistic regression having the lowest, 77.2% and 0.79.

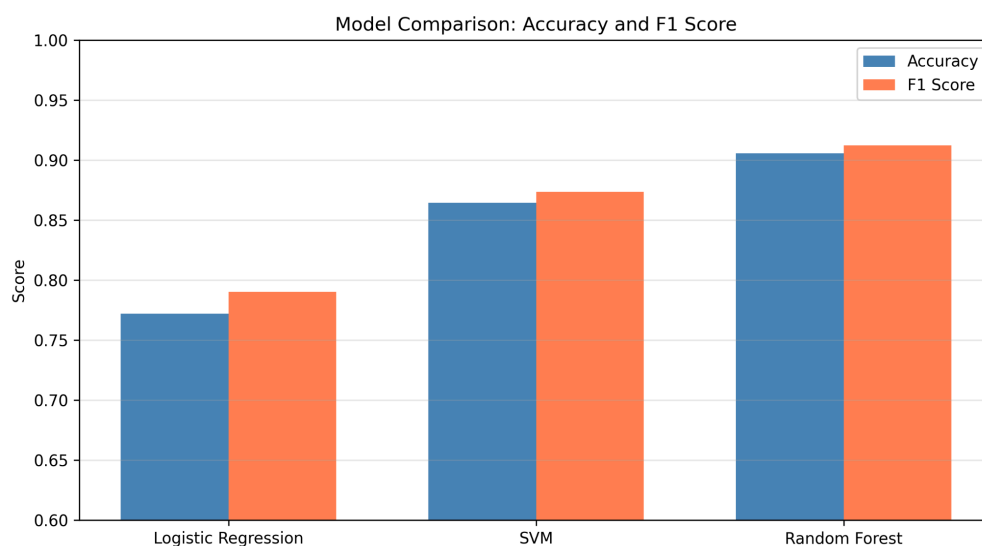


Figure 3. Bar graph showing F1 score and accuracy comparison across three classifiers for binary classification.

The confusion matrix for the highest performing model, random forest, shows 463 true positives, 391 true negatives, 59 false positives, and 30 false negatives.

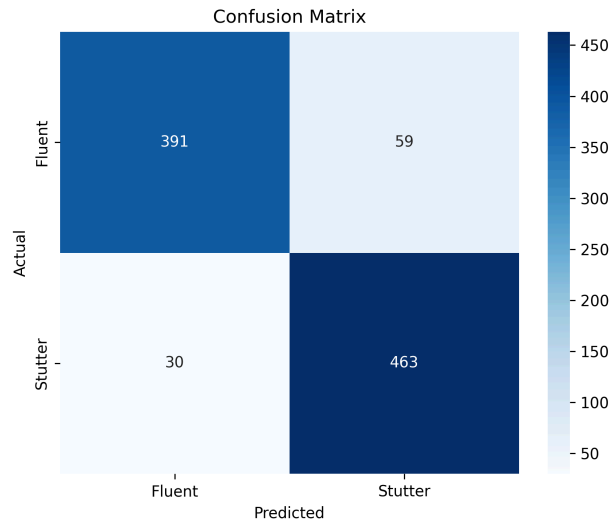


Figure 4. Confusion Matrix showing 463 true positives, 391 true negatives, 59 false positives, and 30 false negatives

Testing for multi-label classification revealed the F1 scores per each category: Block, Prolongation, SoundRep, WordRep, Interjection, and NoStutteredWords. Prior to shuffling, the dataset exhibited ordering bias with the initial five test samples being fluent clips. To assess the impact of shuffling, we evaluated the baseline model on both the original ordered and shuffled splits. The differences were minimal, ranging from 0.01 to 0.08 F1 score across labels, suggesting the original ordering was not severely biased but shuffling was retained as a correct methodological practice to ensure a representative and unbiased test set.

Label	F1 Score before Shuffle	F1 Score After Shuffle
Block	0.7687	0.7439
Prolongation	0.4490	0.3731
SoundRep	0.7296	0.7434
WordRep	0.6552	0.6733
Interjection	0.6625	0.6382

NoStutteredWords	0.8938	0.8952
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Table 1. Per-label F1 scores before and after shuffling for the baseline 13-feature Random Forest model.

The baseline random forest with 13 features led to the resulting F1 scores of 0.74, 0.37, 0.74, 0.67, 0.64, and 0.90. When implementing SMOTE to increase sample sizes for minority categories, the results were 0.35, 0.47, 0.45, 0.67, 0.56, and 0.88. Upon removing SMOTE and increasing the feature count to 29, the results went to 0.78, 0.44, 0.77, 0.74, 0.66, and 0.92. Using the 29 features with a classifier chain led to the F1 scores of 0.79, 0.46, 0.80, 0.72, 0.66, and 0.92.

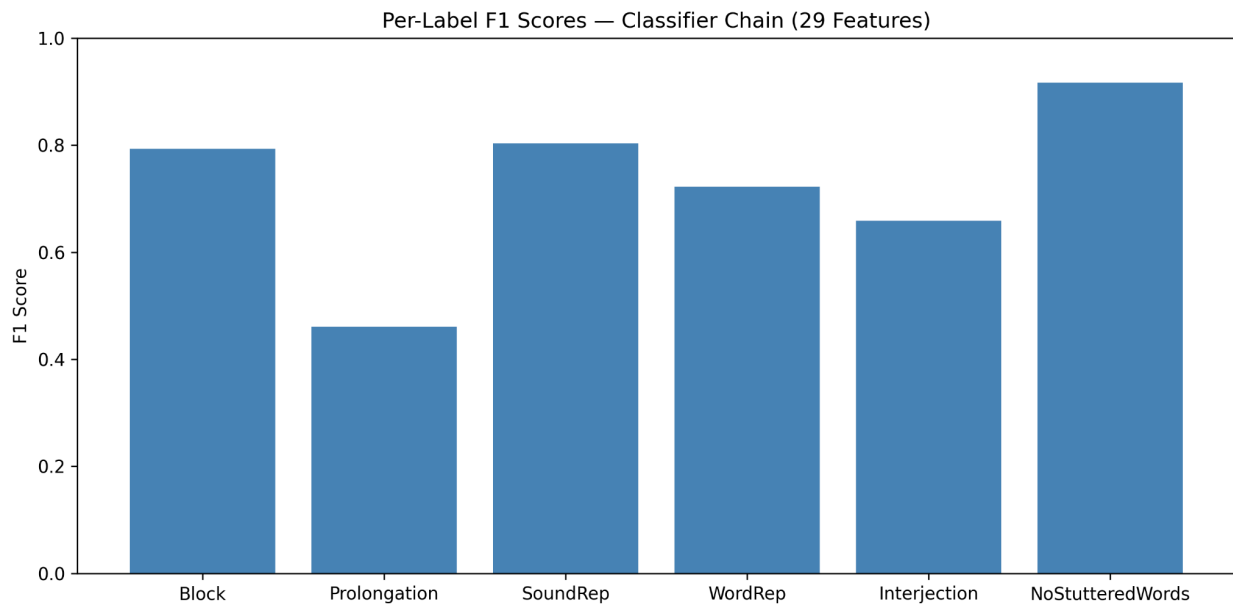


Figure 5. Per-label F1 scores for the best performing multi-label model: the classifier chain with 29 features

Multi-label classifying by using a CNN led to noticeably worse results, achieving F1 scores of 0.69, 0.25, 0.69, 0.06, 0.51, and 0.90. Across all approaches, prolongation was the most difficult to detect, with F1 scores ranging from 0.25 to 0.47.

Discussion:

In the results, random forest outperformed SVM and logistic regression when conducting binary classification. This is due to the fact that random forest can create numerous decision trees to learn patterns, allowing it to be well suited for capturing non-linear patterns, while SVMs can only draw curved boundaries and logistic regression linear ones. The results also show that increasing the number of features from 13 to 29 improved F1 scores. 13 MFCCs could only capture average frequency patterns. When adding standard deviation, the variability of the patterns could be detected, which helps since stutters have much more variability in frequency compared to normal speech. ZCR benefits F1 scores for block detection since blocks have a low crossing rate while they are occurring. Adding RMS helped capture loudness patterns and spectral centroid helped distinguish different sound types. SMOTE hurt the F1 scores overall, and we hypothesize the reason to be that it only generates labels independently of each other, ignoring the fact that multiple labels could occur in one clip. Further investigation will be needed to confirm this reasoning, however. CNN underperformed due to not having enough data samples. With 6.5 million parameters and only 4,712 clips, the CNN overfit the training data instead of learning patterns. Prolongation was consistently the lowest F1 score due to having the fewest positive examples, with only 443 out of 4,712 having a prolongation label. This is consistent with recent work that achieved high accuracy scores after training a CNN on over 30,000 clips (2).

These models were trained and tested on a dataset with clips that were taken in a clinical setting with consistent conditions and audio quality. Model performance on a wide variety of audio sources, such as with background noise, poor audio quality, different accents, different loudness, and different languages remains unknown. Additionally, class imbalances across disfluency types led to lower model performance for those specific categories. Furthermore, the co-occurrence of multiple disfluency types presents a challenge for multi-label classifiers. Future efforts should focus on using larger datasets to allow deep learning methods to perform effectively. It may also be beneficial to use varying kinds of audio sources to

further prepare and train these models so that they could potentially be integrated into speech-language pathology workflows as a tool. Transfer learning using pre-trained audio models, such as Wav2Vec2, could also improve performance, especially for rare disfluency types in this dataset such as prolongation.

References:

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